H

TAMPINES JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

MATHEMATICS 8864/ 01

Paper 1 **Monday, 16 September 2013**

Additional Materials: Answer paper **3 hours**
Graph Paper
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all the work you hand in, including the Cover Page.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A: Pure Mathematics [35 marks]

1 A cold drink is taken from a refrigerator and its temperature, θ °C, t seconds after being out in the open, is given by $\theta = 20 - 15\mathrm{e}^{-0.01t}$. Find

(i) the value of t when $\theta = 15$, [2]

(ii) the value of θ when t becomes very large. [1]

2 The sum of two numbers x and y is 100, and the product of these two numbers is P. Using differentiation, find x and y for which P is a maximum. [4]

3 **(a)** Find $\int \frac{x^3+1}{kx^2}\,\mathrm{d}x$, where k is a non-zero constant. [2]

(b) Differentiate $\ln(x^3+1)$ with respect to x. Hence evaluate $\int_0^2 \frac{kx^2}{x^3+1}\,\mathrm{d}x$, leaving your answer in exact form and in terms of k. [4]

4 The equations of curves C_1 and C_2 are $y = 2\mathrm{e}^{2x} - 1$ and $y = \mathrm{e}^{-x}$ respectively.

(i) Sketch the curves C_1 and C_2 on the same diagram, stating the equations of any asymptotes and the exact coordinates of any intersections with the axes. [3]

(ii) Find the exact area of the region enclosed by the curves C_1 and C_2, the x-axis, and the line $x = 2$. [4]

5 **(i)** Show that $x^2 + 6x + 10$ is always positive for all real values of x. [2]

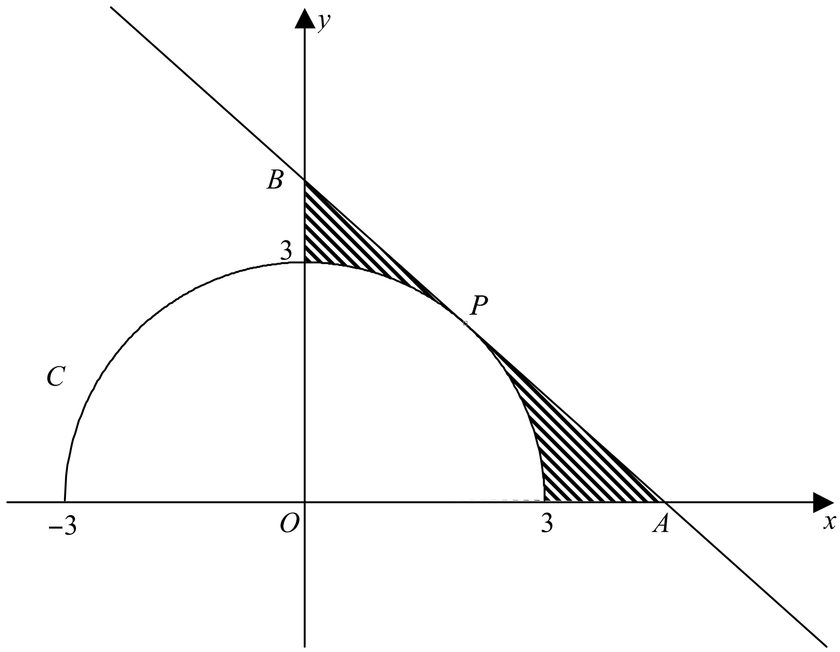
(ii) Solve the inequality $\frac{x^2+6x+10}{x^2+x-6} > 0$. [2]

(iii) Hence solve the inequalities

(a) $\frac{(\ln x)^2 + 6\ln x + 10}{(\ln x)^2 + \ln x - 6} > 0$, [2]

(b) $\frac{x^2-6x+10}{x^2-x-6} > 0$. [2]

6 The semi-circular arc C shown in the diagram below has equation $y = \sqrt{9 - x^2}$. P is a point on C where $x = 2$. The tangent to C at P meets the x-axis and y-axis at A and B respectively.



(i) Find the equation of the tangent, giving your answer in the form $y = \frac{\sqrt{5}}{5}(ax + b)$, where a and b are integers to be determined. [3]

(ii) Hence find the exact coordinates of A and B. [2]

(iii) Deduce the exact area of the shaded region, leaving your answer in terms of π. [2]

Section B: Statistics [60 marks]

7 A recruitment office receives a total of 900 job applications on a particular day. The applications are categorised into the following areas: engineering, business analytics, and social sciences. Due to a shortage of staff, the office is only able to process 100 applications on that day. Hence, the office intends to select a random sample of 100 applications on that day.

(i) Explain what is meant in this context by the term ‘a random sample’. [1]

(ii) Describe how to obtain a systematic sample of size 100 from the job applications. [2]

(iii) Explain why a stratified sample may be preferred in the context of this question. [1]

(iv) Describe how to obtain a stratified sample in the context of this question. [2]

8 Events A and B are such that P(A) = 0.4 and P($A \cup B$) = 0.8.

(i) Showing your working clearly, find P(B) when A and B are

(a) mutually exclusive,

(b) independent. [4]

Assume now that events A and B are independent.

(ii) Using a Venn diagram, or otherwise, find P($A' \cup B$). [2]

For a third event C , it is given that C is a non-empty subset of $A \cap B'$.

(iii) Find an inequality satisfied by P(C). [2]

9 A fruit seller orders 50% of the strawberries from Orchard A, 35% from Orchard B and 15% from Orchard C. The probability that a randomly chosen strawberry supplied by A is bruised is 0.05. The probability that a randomly chosen strawberry supplied by B is bruised is 0.08 and the probability that a randomly chosen strawberry supplied by C is bruised is 0.13.

(i) Show that the probability that a randomly chosen strawberry is bruised is 0.0725. [1]

(ii) Given that a randomly chosen strawberry is bruised, find the probability that the strawberry is supplied by Orchard A. [3]

(iii) Three strawberries are chosen at random. Find the probability that at least two of them are not bruised. [2]

(iv) A customer bought a box containing 100 randomly chosen strawberries from the fruit seller. Using a suitable approximation, find the probability that at least 9 strawberries are bruised. State the mean and variance of the approximation. [4]

10 A technician is monitoring the purity of water. The alkalinity of water x, in milligrams per litre, and the hardness of water y, in milligrams per litre, were recorded over a period of ten days. The results are summarised in the table below.

x	26.2	28.0	27.2	26.1	22.8	33.8	31.8	29.1	31.9	29.4
y	46.0	45.3	43.0	45.0	41.3	51.0	48.0	45.0	50.0	46.3

(i) Draw a sketch of the scatter diagram for the data, as shown on your calculator. [2]

(ii) Find the product moment correlation coefficient and comment on its value in the context of the data. [2]

(iii) Find the equation of the regression line of y on x, in the form $y = mx + c$, giving the values of m and c correct to 2 decimal places. Sketch this line on your scatter diagram. [2]

(iv) Calculate an estimate of the hardness of water when the alkalinity of water is 30 milligrams per litre. Comment on the reliability of this estimate. [2]

(v) It is found that the instrument used to measure the alkalinity of water is faulty and constantly gives a reading of 3 milligrams per litre higher than the correct reading. Without any further calculation, explain how the corrected readings would affect the regression line found in part **(iii)**. [2]

11 A biologist measures the length of a particular species of salmon, chinook salmon, in a fishery to investigate the effect of higher water temperature on the growth of salmon. A random sample of 100 chinook salmon is studied and the lengths per salmon, x mm, are summarized by

$$\sum(x-760) = -150, \qquad \sum(x-760)^2 = 3690.$$

(i) Find unbiased estimates of the population mean and variance. [2]

(ii) What do you understand by the term 'unbiased estimate'? [1]

(iii) The biologist claims that the mean length of a chinook salmon is not 760 mm. Test at the 5% significance level whether the biologist's claim is valid. [5]

The population variance is now known to be 80 mm^2. The biologist suspects that the mean length of a chinook salmon is at least 740 mm. A new random sample of 60 chinook salmon is taken from the same fishery and the mean of this sample is k mm. A test at the 1% significance level indicates that the biologist's suspicion is not justified for this fishery.

(iv) Find the greatest possible value of k , giving your answer correct to the nearest whole number. [5]

12 **(a)** A random variable X is normally distributed with mean 0. Given that $P(X > -k) = p$, where $k > 0$, determine $P(-k < X < k)$ in terms of p, leaving your answer in simplest form. [2]

(b) A factory manufactures two types of tablets, A and B. The mass, in grams, of the tablet of each type have independent normal distributions. The means and standard deviations of these distributions are shown in the following table.

	Mean (g)	Standard deviation (g)
Type A	310	10
Type B	660	30

(i) Find the probability that of 2 randomly chosen tablets of type A, one is heavier than 315 g and the other one is lighter than 305 g. [2]

(ii) Find the probability that the total mass of 6 randomly chosen tablets of type A is within 180 g of the total mass of 3 randomly chosen tablets of type B. State clearly the mean and variance of all distributions that you use. [4]

Tablets of type A are shipped to an exclusive distributor in batches of 10 tablets. The maximum shipping limit allowed before additional cost is incurred for every batch is W g.

(iii) Given that it is 90% certain that any batch of 10 tablets of type A chosen at random has a total mass not exceeding W g, find W, correct to the nearest integer. [3]

(iv) 20 batches are sampled randomly. Find the probability that less than three batches will not meet the maximum shipping limit of W g. [2]

End of Paper